POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2022
Programme: BE | Full Marks: 100
Course: Artificial Intelligence | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is Artificial Intelligence? Explain the current trends of AI and its impact on society. 7

   b) Suppose you are traveling from Pokhara to Kathmandu. You have three agents to travel by flight or by bus or by bike. Suppose your preference is to reach Kathmandu as soon as possible. Which agent do you choose? What is type of this agent? Write the details of task environment of the agent you chose. 8

2. a) What are the constraint Satisfaction Problems? Solve the following Crypt-arithmetic Problem. 8

	B	**A**	**S**	**E**
+	B	A	L	L
G	A	M	E	S

   b) What are the problems of depth first search and breadth first search? How are these problems resolved? Explain with a suitable example 7

3. a) Why is A* search better than best- first search? Explain with a suitable example. 8

   b) Assume the following facts: 7

      i. John likes all kinds of food.
      ii. Apple and vegetable are food
      iii. Anything anyone eats and is not killed is food.
      iv. Anil eats peanuts and still alive
      v. Harry eats everything that Anil eats.
      vi. Anyone who's not killed means, he is alive.

   Represent these facts in predicate logic.

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4. a) Assume the facts in 3 (b) and use resolution to prove "John likes peanuts." 8

b) Represent the following facts in prolog 7

i. John is parent of Pat.

ii. Monica is parent of Pat.

iii. Monica is female.

iv. John is male.

v. Bob is parent of John.

Also write the rules in prolog.

vi. To find who is the father of Pat.

vii. To find who is the grand parent of Pat.

5. a) How does support vector machine work? Illustrate with a suitable example. 8

b) Write down the algorithm for K-means clustering. Explain with a suitable example. 7

6. a) Explain how a single a layer perceptron learn logical OR operation. Assume weights w_1=-0.4 and w_2 = 0.3 and learning rate $\propto$ = 0.21. 8

b) What is Expert System? Explain the architecture of expert system in detail. 7

7. Write short notes on: (**Any two**) 2×

a) Bayesian Network

b) Learning by analogy

c) Phases in Natural language processing

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